

PARC NATL DES PINGUALUIT, QC, Canada

WMO: 719760

Lat: 61.313N

Lon: 73.668W

Elev: 1650

StdP: 13.84

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.9	-36.9	-48.0	0.3	-40.0	-43.9	0.5	-36.5	33.7	-0.2	28.2	-6.8	3.4	250	1.061

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	68.9	56.6	63.5	53.0	59.5	51.7	58.3	66.0	55.2	60.9	52.5	58.1	13.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.6	67.4	60.7	51.7	60.5	57.5	49.4	55.5	55.4	26.0	65.9	24.0	61.0	22.3	58.3	67.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
34.3	30.3	27.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	-16.5	-18.1	-10.4	4.1	21.6	38.3	48.8	47.6	36.5	21.6	6.6	-5.5
	DBStd	25.52	13.07	12.61	11.97	10.41	8.62	9.88	7.15	6.53	6.51	8.31	11.21	11.10
	HDD50	13019	2062	1906	1873	1376	880	376	110	124	408	881	1302	1721
	HDD65	18352	2527	2326	2338	1826	1345	803	506	541	856	1346	1752	2186
	CDD50	148	0	0	0	0	0	27	72	48	1	0	0	0
	CDD65	6	0	0	0	0	0	3	3	0	0	0	0	0
	CDH74	63	0	0	0	0	0	29	34	0	0	0	0	0
	CDH80	14	0	0	0	0	0	7	7	0	0	0	0	0
	WSAvg	12.4	10.7	7.4	11.9	13.3	13.9	12.5	12.6	14.2	14.7	14.0	11.8	11.8
	PrecAvg	17.8	0.7	0.7	1.1	0.9	1.1	1.7	2.0	2.7	2.6	1.9	1.5	1.1
(15) Precipitation	PrecMax	23.8	2.2	2.0	4.1	2.2	2.0	3.3	3.4	5.0	4.6	3.1	2.7	1.9
	PrecMin	12.0	0.2	0.1	0.2	0.2	0.3	0.5	0.5	1.3	1.3	1.0	0.5	0.4
	PrecStd	2.9	0.4	0.5	0.8	0.5	0.4	0.7	0.6	0.9	0.7	0.6	0.6	0.4
	PrecStd	2.9	0.4	0.5	0.8	0.5	0.4	0.7	0.6	0.9	0.7	0.6	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.8	20.4	24.2	32.2	47.0	78.3	79.7	70.4	60.9	40.4	32.5	23.0
		MCWB	29.6	20.0	23.7	32.0	40.7	61.9	61.6	57.8	50.9	39.8	32.3	22.5
	2%	DB	16.2	12.8	17.3	28.5	40.6	66.5	70.7	65.0	52.8	35.9	27.2	18.5
		MCWB	15.9	11.7	16.9	28.3	36.8	53.7	58.4	54.4	47.5	35.4	27.0	18.0
	5%	DB	8.7	5.3	12.3	23.4	37.1	59.4	64.5	60.7	48.9	33.6	24.2	14.9
		MCWB	8.4	4.9	11.8	22.7	34.6	48.9	54.0	53.0	45.8	33.0	23.9	14.4
	10%	DB	3.8	0.8	7.4	20.1	33.7	53.6	60.4	57.4	45.5	31.7	20.5	10.2
		MCWB	3.3	0.2	6.9	19.5	32.5	46.4	51.9	51.4	42.8	31.2	20.1	9.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	17.6	21.6	31.3	39.2	61.3	64.4	60.1	50.5	40.7	30.6	22.8
		MCDB	19.4	17.7	22.1	31.3	42.4	77.1	78.3	65.4	53.4	41.2	30.6	23.1
	2%	WB	10.8	10.0	16.5	27.1	35.6	54.3	59.4	56.5	48.2	35.7	27.1	18.1
		MCDB	11.1	10.9	16.8	27.3	37.8	65.3	69.2	62.6	50.0	35.9	27.3	18.5
	5%	WB	6.4	4.0	11.9	22.1	33.4	49.7	56.2	54.1	44.6	32.9	24.4	14.4
		MCDB	6.7	4.3	12.3	22.6	34.8	58.3	61.8	58.9	47.3	33.1	24.5	14.9
	10%	WB	1.9	-0.7	7.5	18.6	31.3	46.3	53.4	52.0	42.2	30.9	20.8	9.8
		MCDB	2.4	-0.2	8.0	19.1	32.4	53.5	59.7	56.7	44.3	31.3	21.1	10.2
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	10.9	14.0	14.6	11.0	11.7	13.1	10.6	8.9	7.3	10.4	10.8
		MCDBR	15.5	14.5	16.4	17.4	15.5	21.6	20.0	15.6	16.0	7.6	10.4	12.8
		MCWBR	12.7	13.5	14.9	14.9	10.7	10.7	9.1	8.1	10.0	7.9	10.5	12.8
	5% WB	MCDBR	12.1	13.0	15.2	15.0	12.2	20.6	17.6	13.4	13.1	7.5	10.8	12.7
		MCWBR	12.1	13.0	15.1	14.9	10.7	11.0	9.6	8.5	9.9	7.3	10.8	12.7
(40) Clear-Sky Solar Irradiance	taub		0.191	0.233	0.273	0.341	0.341	0.330	0.333	0.325	0.320	0.269	0.202	0.158
	taud		2.034	2.182	2.144	1.959	2.101	2.384	2.470	2.502	2.408	2.260	2.038	1.909
	Ebn at Noon		180	245	269	265	277	285	281	275	252	223	166	135
	Edh at Noon		12	21	31	46	44	34	30	27	25	19	11	8
(44) All-Sky Solar Radiation	RadAvg		117	413	1016	1751	2111	1752	1549	1166	668	256	102	54
	RadStd		8	23	32	62	158	197	195	114	76	37	23	5

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page